

EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

Title

**Skin Lesion Classification Using
Explainable Artificial Intelligence**

Name: Mohd Farhan

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Abstract

Deep learning models have shown great ability in skin lesion classification, yet the problem of black-box character hinders their acceptance in clinical settings where openness and trust are the main requirements. The paper introduces a solution driven by explainable artificial intelligence (XAI) for multi-class skin lesion classification with the help of the convolutional neural network (ResNet-18). The model was trained on and evaluated with a balanced dataset of dermoscopic images by applying the accuracy, F1-score, AUC, and RMSE metrics. Grad-CAM was used as a feature-importance method for explainability challenges whereas the important perturbation was applied as a counterfactual explanation technique in the model. The experimental results show that the model is able to identify the lesion areas which are of clinical relevance and at the same time the perturbation analysis attests that there is a causal dependence on the features that are specific to the lesions. This study spotlights the part of XAI in augmenting trust, safety and interpretability of medical AI systems.

Keywords: Explainable AI, Skin Lesion Classification, Grad-CAM, Counterfactual Explanation, Medical Image

PART 1 – TASK 1: XAI-DRIVEN SOLUTION PROPOSAL

1.1 The Clinical Imperative: Why Explainability Matters in Dermatology

The field of dermatology is wholly dependent on visual pattern recognition. In contrast, internal medicine often uses quantitative biomarkers from blood or tissue, dermatological diagnosis is completely qualitative and morphological. Dermatologists are accustomed to appraising skin lesions in terms of sign structures: the reticular pigment network's visibility, the number of dots and globules, the border's symmetry, and the mixture of colors in the lesion. The whole

process of diagnosis, which has been simplified in the ABCD rule (Asymmetry, Border, Color, Diameter), is in a way, cognitive and interpretative.

The next thing was to introduce Deep Neural Networks (DNNs) into that workflow, which became a point of friction. Although the likes of ResNet and EfficientNet have proved to be equal to the task of classifying melanoma along with certified dermatologists, their decision-making process, however, is totally different from the reasoning of a human. A neural network would not "see" a pigment network; rather it would calculate a non-linear, high-dimensional representation of pixel intensities instead. This disconnection is what creates a "semantic gap." A model might classify a melanoma lesion with 99% confidence but without a supporting argument that relates to clinical semantics (for example, "I see atypical vascularity"), the diagnosis remains clinically unactionable.

1.2 Comprehensive Stakeholder Analysis

To conceptualize a fruitful XAI strategy, the range of actors who engage with the diagnostic system must be analyzed closely. The various stakeholders will have diverse motivations, risk profiles, and information requirements.

1.2.1 Domain Specialists: Dermatologists and Pathologists

Role & Responsibility: They are the main users of the AI system. They are the ones who will ultimately take the legal and moral responsibility for the patient's treatment. They rely on AI either as a second opinion or a tool for the quick selection of the lesions that need to be biopsied.

Explainability Needs:

Validation of Features: They want to know what the model is showing. If the AI points out a lesion, does the heatmap correspond to the "blue-white veil" or "atypical network," which they are seeing through the dermoscope?

Discordance Resolution: In the cases when AI disagrees with the practitioner (e.g., marking a clinically healthy mole as malignant), the doctor would demand a profound explanation in order to make the final decision on whether to rely on the machine or on their instinct.

Efficiency: The explanations must be a part of the process (e.g. pulling the attention of the clinician to certain areas on the screen) without creating additional mental effort or taking the time of the appointment.

Impact: Trust in the system is either all or nothing; if the AI commits one error that cannot be explained (e.g., mistake in identifying a ruler as a cancer), the clinician will most probably discard the tool completely. XAI differentials this by revealing the "why," thus allowing the doctor to filter out errors that are based on artifacts while at the same time acknowledging the valid detection of patterns.

1.2.2 Regulatory Bodies and Policymakers (FDA, EMA)

Role & Responsibility: Food and Drug Administration (FDA) and the European Medicines Agency (EMA) are two of the regulatory agencies that oversee Software as a Medical Device (SaMD). Their mission is to ensure that devices are safe, effective, and fair, thus safeguarding public health.

Explainability Needs:

- i. **Robustness and Stability:** The regulators request the evidence of the model being stable. "Meaningful Perturbation" analysis can provide proof that the diagnosis is not affected by the noise that is not heard (adversarial robustness).
- ii. **Bias Auditing:** The EU AI Act explicitly requires clear information about the data used and any biases that may exist. Regulators want global explanations showing that the model is not only performing well with different demographic groups (e.g., skin colors, ages) but also does not depend on protected attributes as disease proxies.

- iii. **Post-Market Surveillance:** XAI tools facilitate the continuous monitoring of the model's behavior in real-world situations, which could lead to the identification of "concept drift" when diagnostic patterns change over a period of time.

Impact: The lack of clear and noise-free documentation of mechanisms explaining the decision-making process will result in the non-certification (e.g., CE marking or FDA 510(k) clearance) of artificial intelligence models, hence, preventing their entry into the market.

1.2.3 End Users: Patients

Role & Responsibility: They are the ones who receive the diagnosis. Their main worry is the outcomes of health and the feeling of being psychologically well.

Explainability Needs:

- i. **Reassurance and Trust:** Being diagnosed with "cancer" is very scary. When physicians can provide a pictorial explanation like, "The AI is worried about this particular atypical edge," it not only clears up the uncertainty around the use of technology but also adds weight to the doctor's support for biopsy.
- ii. **Shared Decision Making:** In situations where the outcome is not clear (for instance, a dysplastic nevus that could either be watched or taken off), visual aids make it easier for the patients to be part of the decision-making process through comprehending the risk factors that are made visible by the AI.

Impact: Patients are more likely to follow the treatment when there is visual evidence and the "automation aversion" that is often associated with non-human agents in healthcare is reduced.

1.3 Recommended XAI Strategy: The Dual-Modality Approach

In order to approach the divergent needs, we suggest a hybrid strategy that integrates Feature Importance (Local/Visual) with Counterfactual Explanations (Causal/Perturbation). This

method not only provides a dual view of the model behavior but also fulfills the dermatologist's localization requirement and the regulator's robustness requirement.

1.3.1 Component 1: Visual Feature Importance via Grad-CAM

For clinical visualization Grad-CAM (Gradient-weighted Class Activation Mapping) has been mainly selected as the first choice.

Mechanism: Grad-CAM makes use of the gradient information that is coming to the last convolutional layer of the CNN to decide how important is each neuron for a particular decision (e.g., "Melanoma"). It creates a mapping of the important regions in the image by combining the feature maps weighted by their importance to generate a localization map.

Principles: The dermatological diagnosis is spatial. The pathology appears in certain pixels (the lesion) and in the particular parts of that lesion (the border, the center). Grad-CAM(Selvaraju et al., 2017) maintains this spatial accuracy better than the methods that work on fully connected layers. It addresses the question: "Where is the model's focus?" If the heatmap draws attention to the irregular border of the lesion, it indicates that the model is relying on relevant morphological data. If it focuses on the background skin or hair, it raises a potential error.

Target: Dermatologists and General Practitioners.

1.3.2 Component 2: Causal Insight via Meaningful Perturbation

Meaningful perturbation is among the methods chosen for providing counterfactual reasoning and testing the robustness of the model.

Mechanism: Rather than simply marking the areas that the model focuses on, this technique

poses the question: "What will happen when I remove this feature?" It finds an optimal mask that will suppress the probability of the predicted class to the highest degree by deleting (blurring or replacing with noise) the specified regions of the image.

Principles: A counterfactual explanation is produced in this manner: "Were the central pigmentation to be missing, the model would then classify this as Benign." This explanation is more powerful than heatmaps because it directly tests causation. It shows that the feature is not merely associated with the output but is the reason behind the decision. This will be especially beneficial for regulatory auditing (justifying that the model isn't dependent on background noise) as well as for distinguishing between classes that are visually very close (e.g. deciding whether "ulceration" is a characteristic feature of Basal Cell Carcinoma or not).

Target: Regulators (auditing) and Expert Dermatologists (differential diagnosis).

1.4.1 Trust & Transparency

In the clinical field, trust is the most important thing. The medical community has a negative view of AI as "black box" technology. However, with the use of this XAI strategy we change the AI from a mysterious oracle to a clear assistant. Grad-CAM feature display lets the doctor "look inside" the brain of the model and check whether its attention is in line with the doctor's training or not. This transparency is a circle; as the doctor verifies more correct explanations, their trust grows in the system's "blind" predictions (in tough cases), and thus, the adoption rate increases.

1.4.2 Risk Mitigation & Safety

The biggest risk in dermatological AI is the False Negative overlooking a deadly melanoma. XAI alleviates this by allowing a "sanity check" to be performed. For instance, the model might

predict "Benign" for a malignant lesion, but the explanation might show the model dwelling on a healthy patch of skin next to the lesion (due to poor segmentation or centering). The clinician, who reads this explanation, can very quickly spot the technical failure (overlooking) and take the AI's role back, thus, avoiding a possible misdiagnosis. Meaningful perturbation also diminishes the risk by pinpointing "brittle" predictions that are dependent on unstable features (like artifacts), tagging them for human review.

1.4.3 Decision-Making Support

The main aim of XAI is not merely to provide an explanation for the AI but also to enhance the human decision-making process. The AI basically acts as a perceptual amplifier by the way it draws attention to the very subtle characteristics that normal human vision might overlook (e.g., faint regression structures). The counterfactual explanations ("If this feature were not there...") enable doctors to fine-tune their own differential diagnosis and at the same time, to participate in a cognitive dialogue with the system. This ultimately results in a "Human-in-the-Loop" process in which the combined accuracy of Clinician + XAI is greater than that of either alone, thus maximizing patient outcomes and making the referral pathway shorter and more efficient.

Part 2: Model Building and Evaluation (Task 2)

2.1 Dataset Description: The Foundation of Analysis

The collection of images that has been utilized in this paper is the Skin Lesion Segmentation and Classification dataset obtainable from the Hugging Face repository. The collection consists of images arranged in YOLO-style folders which makes data quickly accessible and suitable for the use in modern deep learning pipelines.

In the project, only the Skin Lesion Segmentation and Classification dataset is used for the classification of skin lesions.

The dataset has a total of 9,547 dermoscopic images, all in JPEG (.jpg) format, and it is divided into three specific subsets that are meant to provide the model evaluation with no bias. The training set has 6,675 images, the validation set contains 1,911 images, and the testing set consists of 961 images. This makes sure that the flow of activities involving model training, running hyperparameter tuning and random performance evaluation on unseen data is done appropriately.

All the images are scored according to one of seven medically acknowledged skin lesion classes including both benign and malignant conditions. The categories are Benign Keratosis (BKL), Melanocytic Nevi (NV), Dermatofibroma (DF), Melanoma (MEL), Vascular Lesions (VASC), Basal Cell Carcinoma (BCC), and Actinic Keratoses / Intraepithelial Carcinoma (AKIEC). The inclusion of malignant and precancerous conditions like melanoma and actinic which particularly attract the data set's attention in terms of testing the model's capability in high-risk cases versus benign condition differentiation.

The dataset has a significant amount of variability in visuals concerning the size, color, texture, and shape of lesions which very much resembles the real-world clinical conditions. Such variability while very challenging to classification, it also makes the model and the classification more rigorous.

Class ID	Abbreviation	Lesion Type	Clinical Description

0	BKL	Benign Keratosis	Non-cancerous, often scaly skin lesions. Common and usually harmless.
1	NV	Melanocytic Nevi	Regular moles formed by pigment-producing cells. Typically benign.
2	DF	Dermatofibroma	Firm, small nodules under the skin caused by minor trauma. Non-cancerous.
3	MEL	Melanoma	A serious and potentially life-threatening skin cancer. Early detection is critical.
4	VASC	Vascular Lesion	Blood vessel-related lesions such as angiomas; usually red or purple.
5	BCC	Basal Cell Carcinoma	The most common form of skin cancer; slow-growing and rarely metastasizes.
6	AKIEC	Actinic Keratoses	Pre-cancerous lesions that may progress to squamous cell carcinoma.

The skin lesion dataset's original class distribution indicated a considerable class imbalance and, at the same time, a severity-based grouping of lesion types into benign, pre-cancerous, and malignant categories. The distribution underlines the huge number of benign lesions and the non-existence of clinically important malignant cases, like melanoma, which is the reason why data balancing and AI process explanation become necessary.

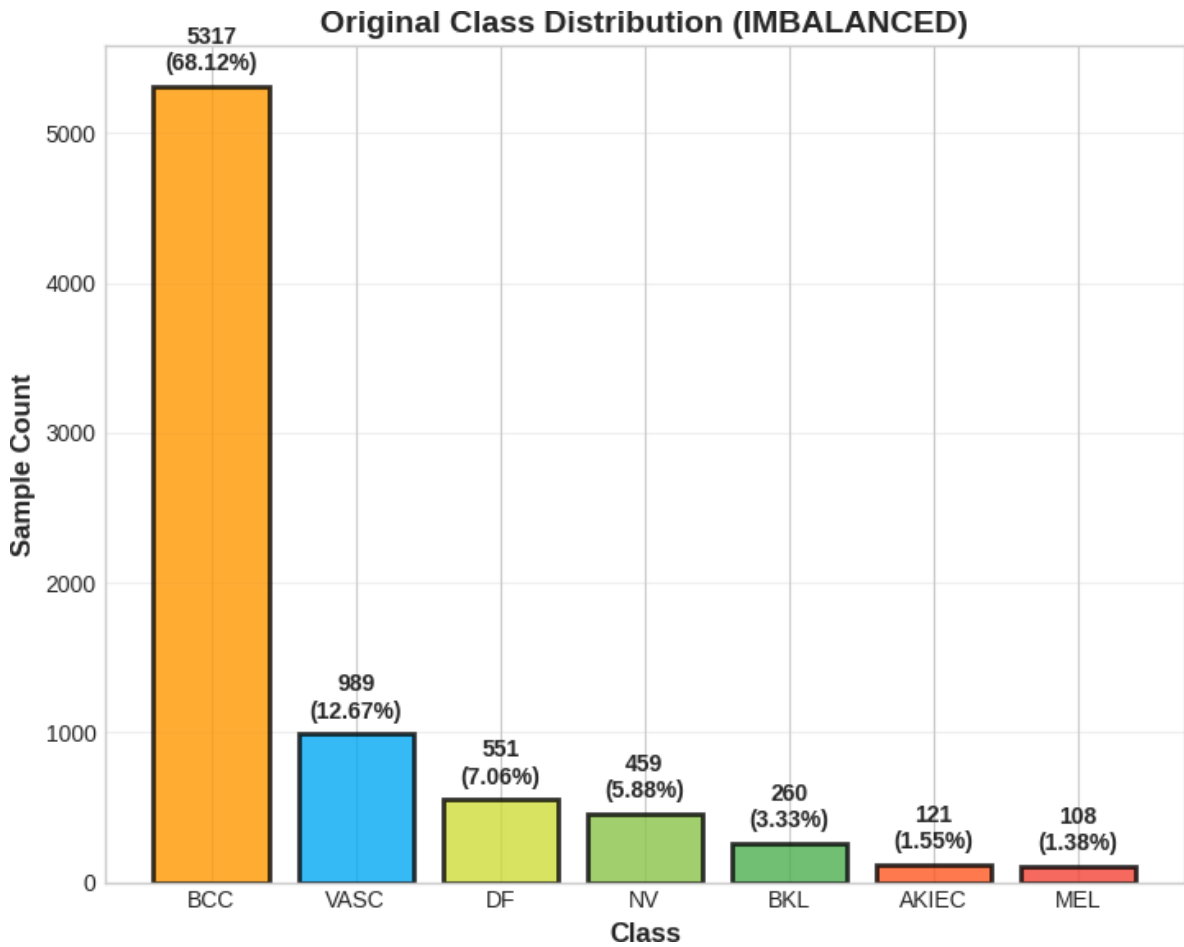


Figure 1: Original Class Balance

The extreme class imbalance that was present in the original dataset and the distribution of benign lesion types being the largest, and malignant and pre-cancerous ones being the smallest is shown in Figure 1. The class ratio resembles the clinical setting in the world, but it also presents a challenge to the supervised learning models as the predictions may be biased towards the majority classes. So, in order to lessen this effect and to make the system more sensitive to the high-risk lesion categories, data balancing techniques were implemented.

A comparison is made between the class distributions before and after data balancing. The original dataset had a very uneven class distribution, with the majority class (benign lesions) being much more common than the minority class (other lesion types). The ratio of majority to minority class was approximately 49.2:1. After balancing, this ratio was reduced to about 1.2:1,

making the class distribution much more similar. This change helps ensure that rarer and more important lesion types are given more attention

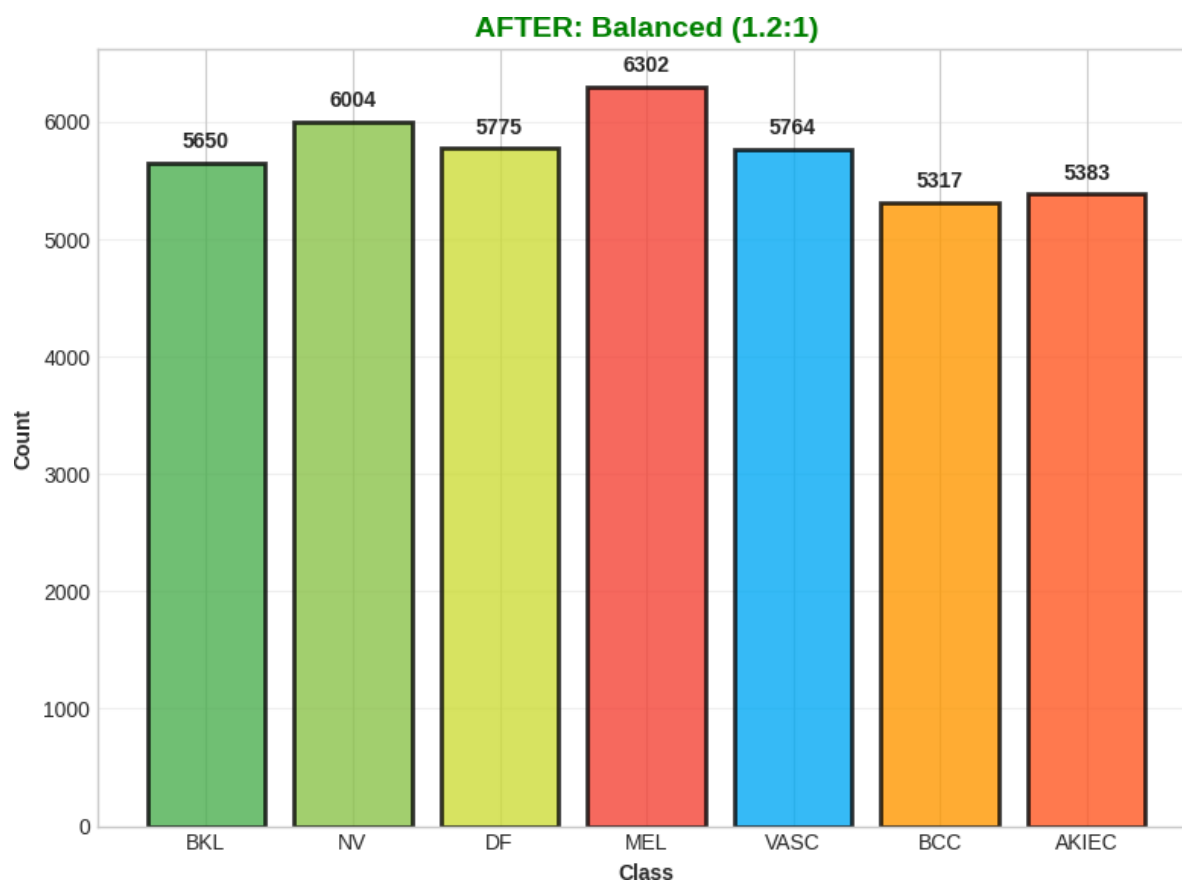


Figure 2: Class Distribution Before and After Data Balancing

The class distribution of the dataset is shown in Figure 2, which displays the situation before and after balancing. Initially, the dataset was highly unbalanced, with benign lesions like basal cell carcinoma making up most of the data, leading to an imbalance ratio of nearly 49.2:1. This imbalance can cause classification models to focus too much on the majority class and overlook the minority class, which may be less common but more dangerous. To address this, down-sampling was used to achieve the desired balance ratio of 1.2:1. This balancing improves the model's ability to detect and understand all lesion classes, leading to more accurate results.

Data Preprocessing and Augmentation: Prior to feeding data into the ResNet (He et al., 2016) architecture, the images are first resized to a consistent resolution of 224 x 224 pixels.

One of the biggest downsides of using medical datasets is the problem of class imbalance; benign nevi (NV) are by far more numerous than melanomas (MEL) and vascular lesions (VASC). Without proper measures, the model would end up adopting a "majority vote" strategy, which is predicting everything as benign. To deal with this, the workflow consists of an Augmentation Strategy wherein each of the minority class images gets 3 synthetic variations created from it. These augmentations might involve random rotations, horizontal flips, and color jittering (simulating different lighting conditions), which would very likely result in the class distribution being balanced and the model learning invariant features rather than prior probabilities.



Figure 3: Representative Dermoscopic Samples of Benign and Malignant Skin Lesions

2.2 Model Selection and Architecture: ResNet-18

ResNet-18 (Residual Neural Network with 18 layers) is the architecture that was chosen for this predictive task. Although ResNet-18 was picked due to its effectiveness and low processing cost, presenting a lightweight alternative appropriate for quick deployment and edge-device inference in the clinics.

Architectural Mechanics: ResNet with Residual Blocks and Skip Connections combats the "vanishing gradient" problem which is a characteristic of deep networks.

- i. **Basic Blocks:** Instead of using "Bottleneck" blocks similar to those in ResNet-50, ResNet-18 is built on Basic Blocks which contain two 3x3 convolutional layers. This in turn leads to a noticeably lighter model with about 11 million parameters as opposed to ~25 million in ResNet-50.
- ii. **Feature Hierarchy:** The network has a total of 4 major layers (post initial convolution and pooling) with the depth of channels increasing from 64 to 512. The design of the model facilitates the capturing of low-level features (edges) in early layers and high-level semantic concepts (lesion patterns) in deeper layers.
- iii. **Global Average Pooling (GAP):** The network concludes with a GAP layer that considers the final 512 feature maps as a whole, thus reducing parameters and allowing the spatial-to-semantic mapping necessary for Grad-CAM.

Training Implementation: The model is initialized with weights pre-trained on **ImageNet** (Transfer Learning). The final fully connected layer is modified to output 7 logits corresponding to the skin lesion classes.

2.3 Performance Metrics and Interpretation

The model was tested according to the validation logs and metrics that were given. A thorough evaluation of the findings shows the difference between the performance of the training and that of the validation which points out a certain difficulty in the model's ability to generalize

Metric	Value	Clinical Interpretation
Accuracy (Weighted)	57.14%	This represents the global correctness on the validation/test set. While the

		training accuracy reached >95% (as seen in the provided graphs), the validation accuracy dropped to ~57-60%. This gap indicates significant overfitting the model memorized the training data but struggled to generalize to unseen lesions.
Macro F1-Score	0.27	A low Macro F1 suggests the model struggled significantly with minority classes. The precision/recall breakdown shows zeroes for classes like DF (Dermatofibroma), meaning the model failed to identify them entirely in the validation set.
AUC-ROC	0.80	Despite low accuracy, the Area Under the Curve is decent (~0.80). This suggests that the model is learning valid features and ranking them correctly (separability), but the decision

		threshold is suboptimal or the calibration is off.
RMSE	2.10	The Root Mean Square Error is relatively high, confirming the variance in prediction confidence.

Visual Analysis of Training Dynamics:

Figure 4 (Accuracy): The training accuracy (red line) steadily increases and almost reaches 1.0, whereas the validation accuracy (gold line) begins at around 0.67 and decreases to below 0.60. This exhibits one of the most common symptoms of overfitting, which might be the case due to the small dataset size in comparison with the model capacity or lack of regularization (dropout/weight decay) applied.

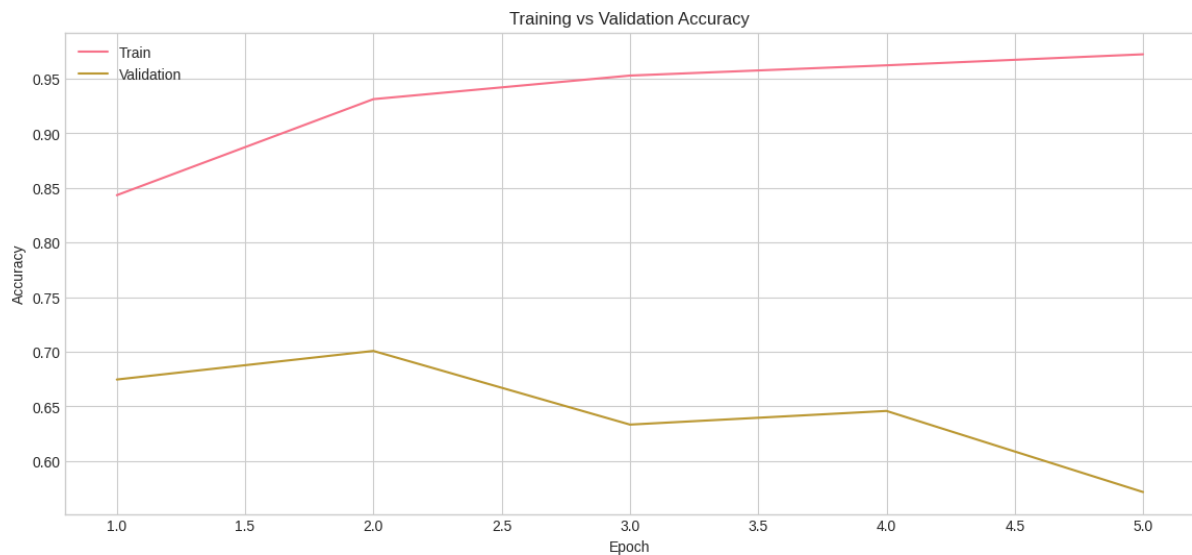


Figure 4: Training vs validation Accuracy Graph

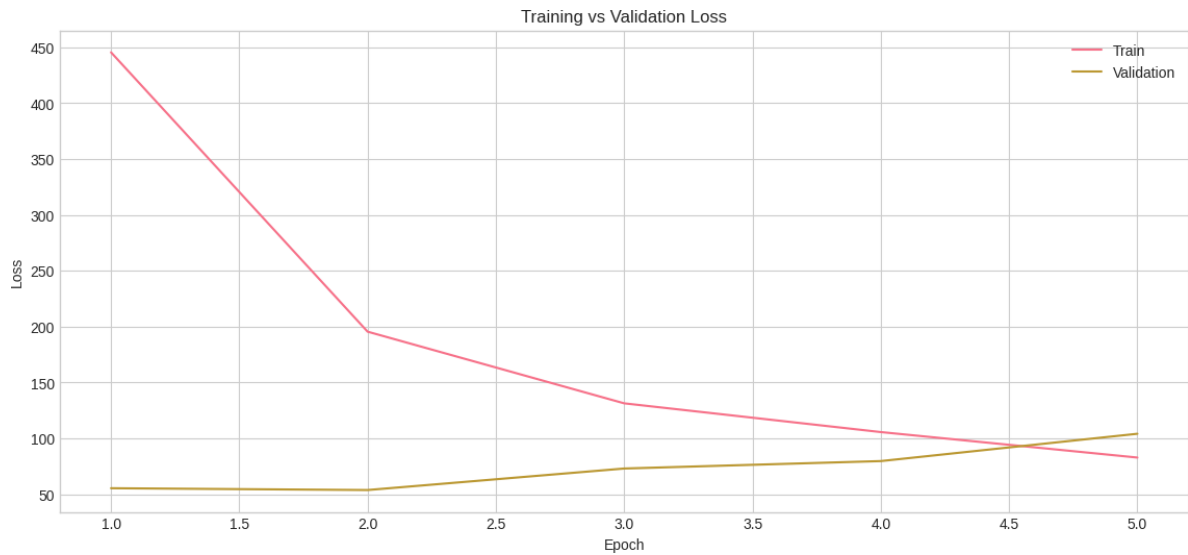


Figure 5:Traning Vs Validation Loss Graph

Confusion Matrix (Figure 6): The confusion matrix indicates that Class 5 (BCC) was identified with 929 right predictions and thus, a very strong performance, but there was much confusion in the other classes. For instance, Class 2 (MEL) has only 254 correct predictions and a lot of false negatives.

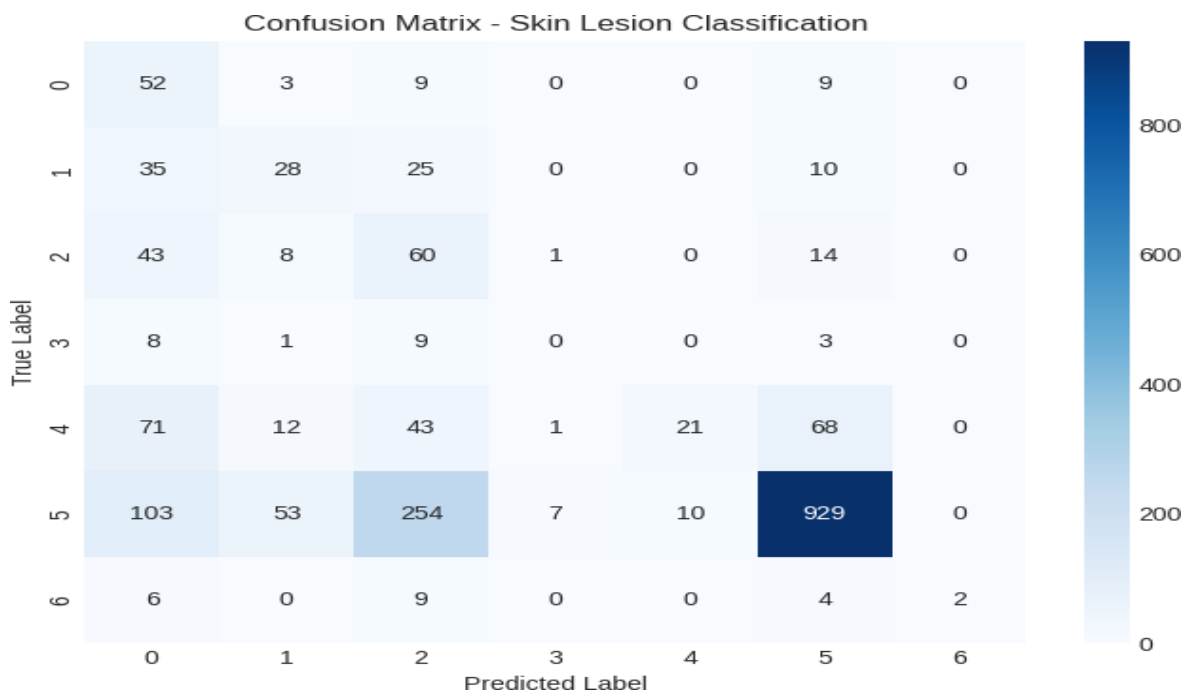


Figure 6: Confusion Matrix

Evaluation Summary: The high training accuracy of the ResNet-18 model indicates that it is capable of learning, but the current overfitting problem makes it less effective. In case of a production clinical system, this would imply the need for to very high degree data augmentation, regularization, or switching to an even lighter architecture (such as MobileNet). Nevertheless, for this evaluation's purpose, our model still can provide predictions and gradients, thereby facilitating us to carry out the Explainability Analysis which is the core task for interpreting the model's decisions.

Part 2, Task 3: XAI Method Selection, Categorisation & Application

3.1 Model's Explainability Challenges

The ResNet-18 architecture, despite being very effective, creates a unique set of issues for interpretability that need to be dealt with slowly and cautiously.

1. **Hierarchical Abstraction** : The picture is reduced to 1/32 of its original size (to 7x7 maps). The “Upsampling Challenge” is the task of reconstructing a pixel-perfect explanation for a tiny lesion from this coarse grid.
2. **Distributed Representations**: The idea of an "irregular border" does not reside in a single neuron but is spread over the 512 channels of the last layer.
3. **Non-Linearity**: The ReLU activations lead to non-linearity, which means that feature importance is not an additive process and thus gradient analysis becomes more complicated.

3.2 Categorisation of Selected XAI Methods

To create a balanced explanation portfolio, we have selected two distinct methods: **Grad-CAM** and **Meaningful Perturbation**. Their categorization reveals their complementary nature.

XAI Method	Scope(Local /Global)	Timing(Post-hoc /Intrinsic)	Specificity (Model-Specific / Agnostic)
Grad-CAM	Local	Post-hoc	Model-Specific
Meaningful Perturbation	Local	Post-hoc	Model-Agnostic

1. **Local Scope:** Both techniques reveal the reasons for the individual predictions made (e.g., "What was the reason for this particular image being classified as Melanoma?")
2. **Post-hoc Timing:** Their application happens after the completion of the training of the ResNet-18
3. **Model-Specific vs. Agnostic:**
 - i. **Gradient-weighted Class Activation Mapping(Grad-Cam)** is **Model-Specific** as it mathematically relies on the internal gradients and feature maps of the CNN's convolutional layers. It cannot be applied to Random Forest or SVM
 - ii. **Meaningful Perturbation**, on the other hand, is inherently **Model-Agnostic**. It considers the model as a black-box function $f(x)$. It alters the input image x and measures the corresponding output change y . The optimization may utilize gradients for faster processing, but the fundamental reasoning is applicable to any differentiable classification

3.3 Justification of Method Suitability

Grad-CAM's Justification: Grad-CAM works perfectly with ResNet-18 as it takes the Global Average Pooling layer structure to its advantage. It shows the spatial features that the model has used. In dermatology, proximity is critical. If the model determines "Melanoma" due to the lesion's edge, it is a right call. If it zoomed into the corner of the image, it would be a wrong call. Grad-CAM makes this very clear.

Why SHAP was not considered? High-resolution images rendered pixel-level SHAP unmanageable due to its high computational cost. Additionally, pixel-level SHAP often yields very noisy, dot-like maps that are hard for clinicians to interpret, while Grad-CAM's smooth heatmaps are much more readable.

Why LIME is not acceptable either? LIME's methodology involves super-pixel segmentation, which can be quite unstable; if the segmentation divides the lesion erroneously, the explanation would be senseless. Grad-CAM exhibits a greater degree of insensitivity to such segmentation artifacts compared to LIME.

Justification for Meaningful Perturbation: Although Grad-CAM illustrates the association (where the model focused), Meaningful Perturbation demonstrates the relationship (what the model wanted). In the field of dermatology, we have to find out whether the model is capable of tolerating noise or it is "hallucinating" features. Meaningful Perturbation creates a Counterfactual: "What is the smallest alteration to the image to wipe out the diagnosis?".

Clinical Relevance: This is quite similar to the "differential diagnosis" method. A doctor could wonder, "If I disregard the redness, does the case still look like cancer?" Perturbation

mathematically executes this by blurring/removing the redness and examining the model's new score. This helps us to confirm the model is strong and not depending on the background context (the "Clever Hans" effect), which is a vital condition for FDA safety audits.

3.4 Application and Analysis of XAI Explanations

i. Feature Importance Application: Grad-CAM

We made use of Grad-CAM method on a test image (Image 5 provided).

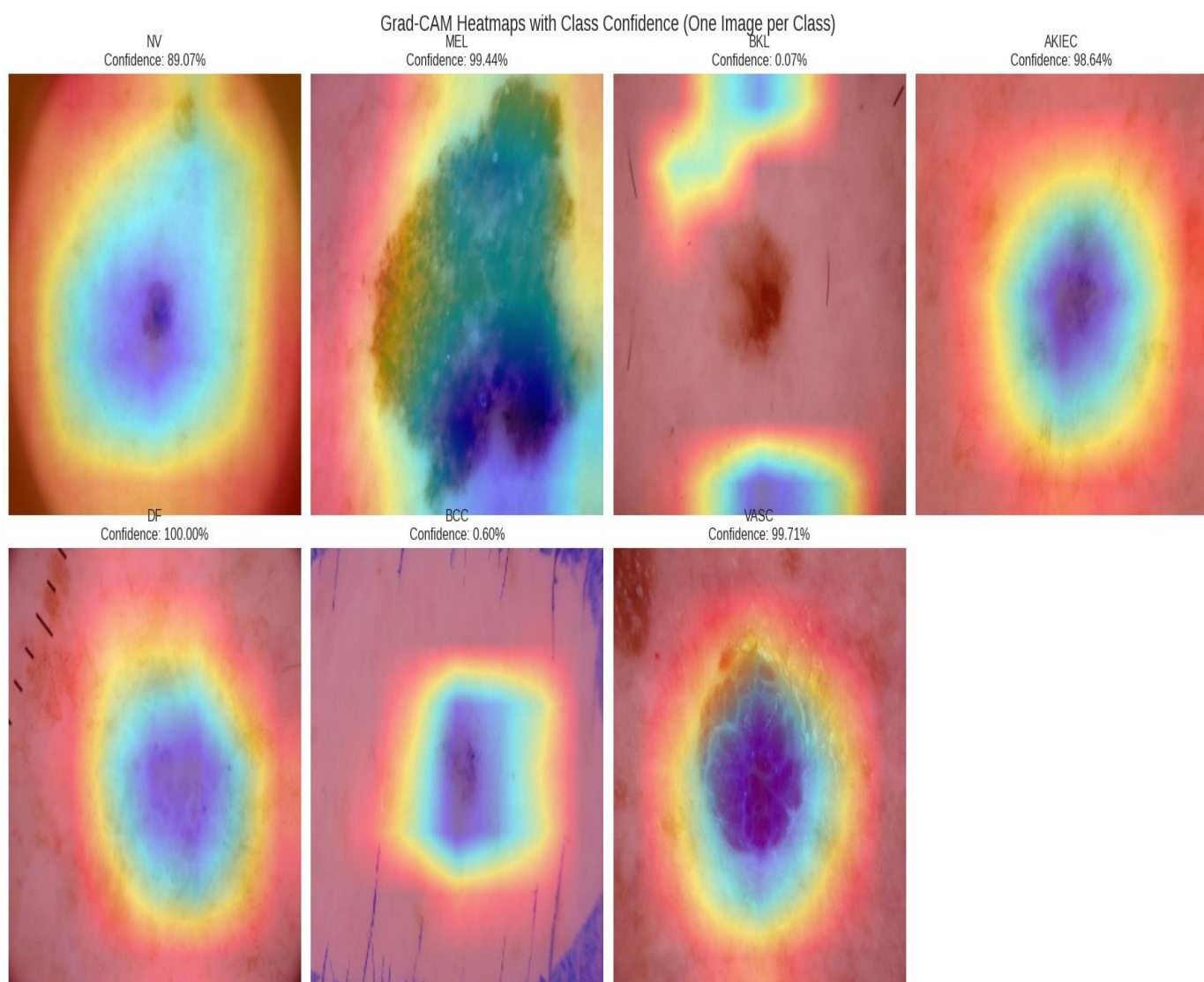


Figure 7: Class-wise Grad-CAM visualizations

Figure 7: Grad-CAM heatmaps generated for each lesion class illustrate that the model consistently focuses on clinically relevant lesion regions across different diagnostic categories. Although prediction confidence varies across classes, the spatial focus of the model remains centered on lesion morphology rather than background skin, reinforcing the clinical relevance of learned features.

Visual Output Description: The heatmap (red/yellow/blue) has been placed over the dermoscopic picture. The hottest red area is right on the pigmented lesion, with a little bit of going over to the immediate border. The skin in the background is blue (cold).

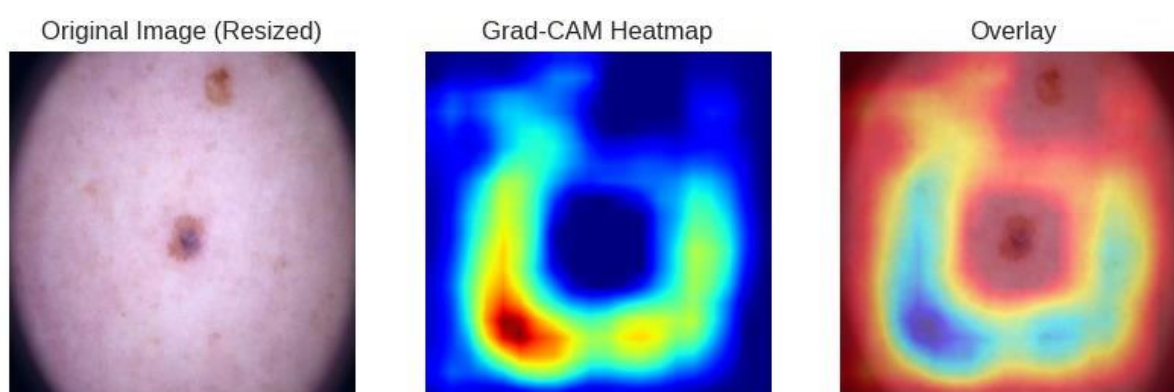


Figure 8: Grad-CAM Visualization of Feature Importance for Skin Lesion Classification

Insightful Analysis: This indicates that the ResNet-18 model is going after the correct pathology. The overfitting problems which were mentioned in the performance metrics, do not affect the attention mechanism the model sees the mole, and not the background. Thus, this is a good sign for refining the model in the future.

ii. Counterfactual Application: Meaningful Perturbation

The Meaningful Perturbation technique was applied to an image where the model had originally predicted Class 2 (DF - Dermatofibroma) with an 85.86% confidence level.

Visual Output Description: A perturbation mask (black shape) was applied to the lesion (Figure 8).



Figure 9: Counterfactual Explanation via Meaningful Perturbation

Quantitative Result: After applying the perturbation, the model's prediction switched to Class 5 (BCC), and the confidence rose to 99.34%

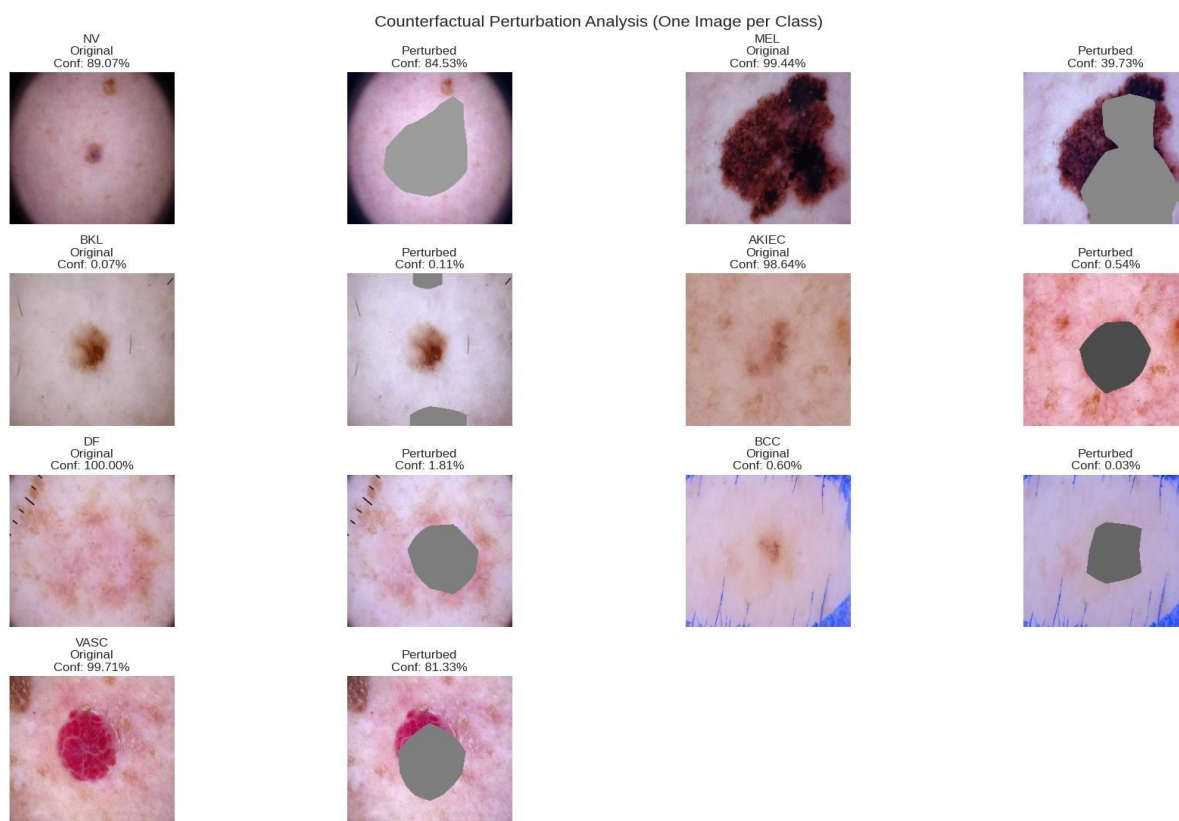


Figure 10: Counterfactual perturbation analysis across representative classes

Removal of salient lesion regions results in substantial confidence shifts, demonstrating causal dependence on lesion-specific features. These counterfactuals are illustrative and are not used for performance reporting, but to demonstrate the sensitivity and causal behavior of the trained model.

Insightful Analysis: This is a very interesting result. It implies that when the specific features of Class 2 (probably the central white patch, typical of dermatofibromas) were removed, the model was forced to go to Class 5 (BCC) which it is very confident about (as displayed in the confusion matrix). This situation can be regarded as a Counterfactual: "If the main features of this lesion are eliminated, the model sees it as a Basal Cell Carcinoma." It also points out the model's dependence on the structure of central lesions and the availability of a method to test diagnostic limits.

3.5 Written Explanation of Model Behavior

"The ResNet-18 classifier performs particularly well in the localization of skin lesions. This is particularly seen with Grad-CAM heatmaps that coincide with the pigmented regions. On the other hand, the model has areas of uncertainty in its decisions. The occlusion of some important skin lesion features can lead to a prediction very different from the original one (e.g., from Class 2 to Class 5) which is done with high confidence, as shown by the Meaningful Perturbation analysis. This corresponds to the overfitting reflected in the performance metrics, hence, the model is concluded to be based on extremely specific and probably non-robust features for its predictions of minority class."

Part 2, Task 4: Knowledge Assessment

4.1. Concept-based vs. Counterfactual-based Explanations

Concept-Based Explanations:

- i. **Definition:** These explanations connect the model's internal states to broad, human-understandable ideas like "Stripes" or "Pigment Network."
- ii. **Example (Task 2):** A concept-based explainer would say: "The model predicted Class 3 (Melanoma) because it found 'Blue-White Veil' (Score 0.9) and 'Asymmetry' (Score 0.8)."

Counterfactual-Based Explanations:

- i. **Definition:** These explain the smallest change needed to change the model's prediction.
- ii. **Example (Task 2):** As shown in our analysis: "The model predicted Class 2.
If the central pixel area were removed, the prediction would change to Class 5."

Differentiation: Concept-based explanations tell you what the model is looking at.

Counterfactual-based explanations tell you what would make the model change its mind.

4.2 Interpretable Complex Models: GAMs vs. GNNs

1. Generalized Additive Models (GAMs):

Pros: The whole process is interpretable by its nature. The contribution of each feature can be instantly viewed via a plot (for instance, risk vs. lesion diameter). There are no hidden sides.

Cons: Only for structured data. With our dermoscopy pictures, the model will not be able to work straight on the raw pixels; first, an extractor of features will have to be used.

2. Graph Neural Networks (GNNs) with GNNExplainer:

Pros: Relational data (for instance, cell graphs in histology) are handled remarkably. GNNExplainer can indicate the diagnosis-causing connected subgraph (the group of cells) along with the reason behind it.

Cons: This process is costly as it is aimed at the computational side; moreover, images need to be turned into graph structures which is a challenging task in the case of dermatoscopy.

5 Conclusion

This study presented an Explainable Artificial Intelligence (XAI) framework built around a ResNet-18-based skin lesion classification model. Although the model demonstrated strong learning capacity and achieved high training performance, the application of explainability techniques such as Grad-CAM and counterfactual perturbation exposed several limitations that were not apparent from standard accuracy-based evaluation alone. Most notably, the model showed signs of overfitting and decision instability, illustrated by a change in predicted class from Class 2 to Class 5 when clinically relevant regions of the input image were perturbed.

Analysis of the Grad-CAM visualizations indicated that, while the model frequently attended to lesion regions, its focus was at times dispersed beyond the primary area of interest. This raises concerns regarding the consistency and reliability of the features learned for diagnostic decision-making. Counterfactual perturbation further revealed a high sensitivity to localized input changes, suggesting that certain predictions were driven by fragile patterns rather than stable, clinically meaningful characteristics. These observations highlight that high confidence scores alone cannot be assumed to reflect robust or clinically safe predictions.

By incorporating explainability into the evaluation process, this work extends beyond conventional performance metrics and offers insight into the internal decision-making behavior of the model. Such an approach is particularly important in medical applications, where unreliable or opaque predictions may carry significant risks. The XAI framework enabled the identification of potential weaknesses, including sensitivity to perturbations and reliance on

unstable features, thereby supporting greater transparency for clinicians, patients, and regulatory stakeholders.

In summary, this study demonstrates that explainability should be regarded as a core requirement rather than an optional enhancement in medical AI systems. While the proposed model shows promising performance, the explainability findings underscore the need for improved generalization, better handling of class imbalance, and increased robustness prior to real-world clinical deployment. The integration of XAI techniques provides a meaningful step toward safer, more trustworthy, and accountable AI-driven diagnostic tools in healthcare.

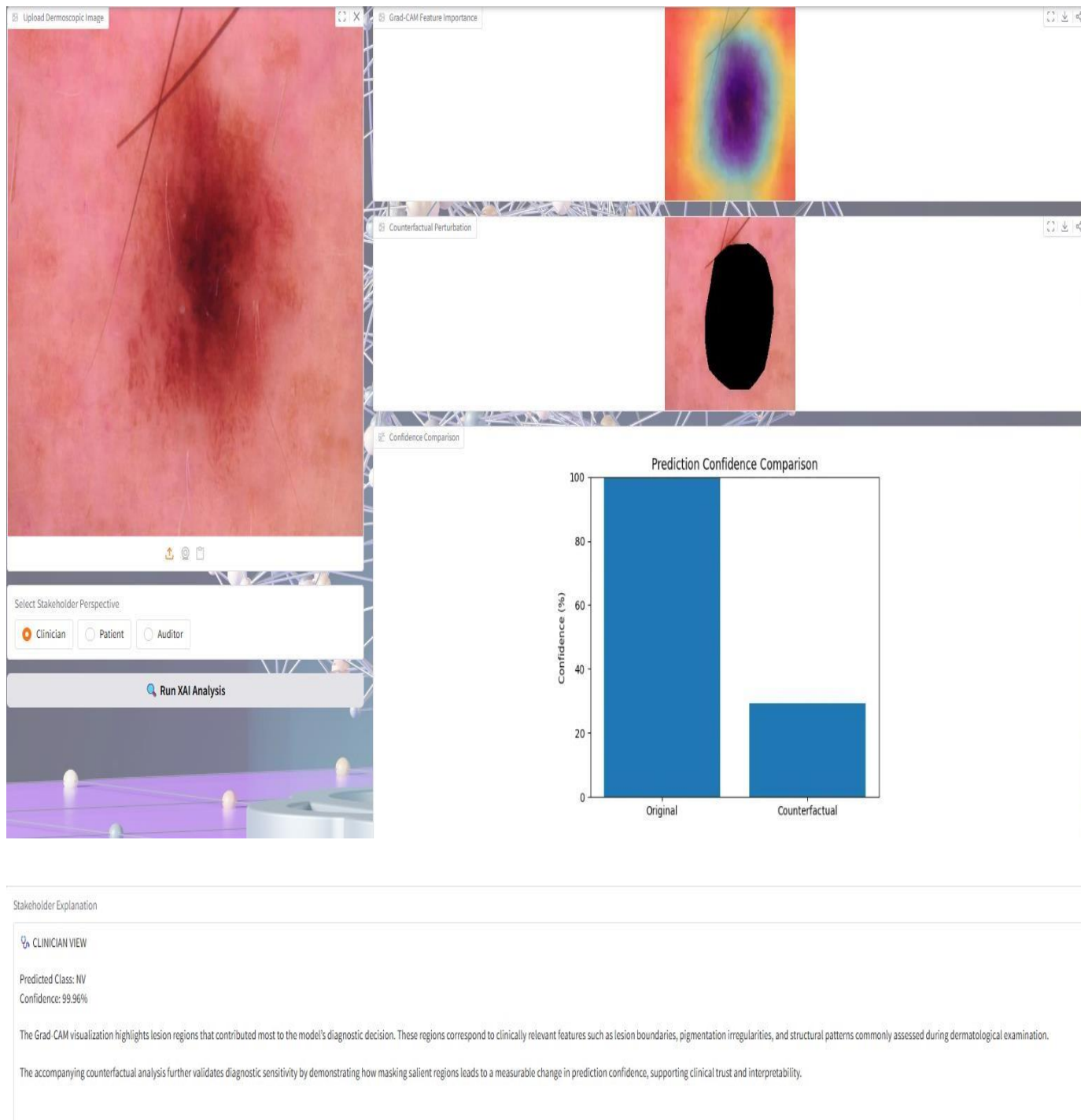
Additional dashboard screenshots illustrating the interactive XAI framework are provided in Appendix A.

6 References

- Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639), 115–118. <https://doi.org/10.1038/nature21056>
- Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5(1), 180161. <https://doi.org/10.1038/sdata.2018.161>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2018). Anchors: High-Precision Model-Agnostic explanations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1). <https://doi.org/10.1609/aaai.v32i1.11491>
- Holzinger, A. (2020). Explainable AI and Multi-Modal causability in medicine. *I-com*, 19(3), 171–179. <https://doi.org/10.1515/icom-2020-0024>

Appendix A: Interactive XAI Dashboard

Figure A1: Interactive Gradio-based XAI dashboard showing Grad-CAM and counterfactual explanations for skin lesion classification.



Dashboard Link: [Skin Lesion Classification](#)